Map Skills & Atlases

1. Using an Atlas index

An Atlas index - An alphabetical list at the back of an Atlas and helps you locate any place quickly.

It gives you:

- In Alphabetical order : The places listed from A-Z (e.g. Cape Town is before Durban).
- The Page Number: It tells you the exact page to turn to.
- The Grid Reference: Gives you the letter and number (e.g. B3) to find the specific box on the Page.
- Latitude and Longitude: Gives you the exact global coordinates in degrees and minutes (e.g. 33°55'S, 18° 25' E)

How to read an Index Entry

Example:

Durban..... 24 B3..... 2

- *Place Name: Durban*
- *Atlas page: Page 24*
- *Alpha-numeric grid: Grid block B3*
- *Coordinates: 29°53' S (South), 31°03' E (East)*

2. Map Symbols and Keys (Legends)

Maps use small pictures, colours, and shapes called **symbols** represent real-world features without overcrowding the map.

- Map Key/Legend: A box on the side or bottom of the map that explains what each symbol means.
- Point Symbols: They mark single, specific locations (e.g. a dot for a city, a cross for a hospital, or a small plane for the airport)
- Line Symbols: Show us the linear features that connect the places(e.g. roads, rivers, railway lines, borders)
- Area Symbols: Use colours to show the regions covering the ground (e.g. green or forests, blue for lakes or oceans).

3. Map Scales

A map scale shows the relationship between a distance measured on the map and the actual distance on the ground.

Types of Scales:

1. Word Scale: Expressed in plain words.
- Example: "One cm(centimeter) on the map represents one kilometer on the ground."

2. Line Scale: A line drawn like a ruler marked with real ground distances. You measure distance on the map with a ruler or piece of paper and line it up against the scale to get the real distance.

3. Ratio Scale: It is expressed as a ratio where both numbers use the same unit (usually in centimeters).
- Example: **1:50 000**, which means 1 cm on the map equals to 50 000cm (500m) on the ground.

4. Grid Systems (Alpha-Numeric Grids)

Alpha-numeric grids divide a map into horizontal rows and vertical columns to locate places quickly on local or regional maps.

- **Columns:** Usually labeled with letters (A, B, C, D,...) across the top or bottom.
- **Rows:** Usually labeled with numbers (1, 2, 3, 4,...) down the sides.
- **Reading order:** Always read the **letter first** then the **Number** (e.g. C4)